Harshith Gowda Shakaladevanapura Maregowda

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Research Interests

Machine Learning for dynamic systems, Reinforcement Learning, Data-driven control, Process systems modeling, Experimental system identification, Real-time control implementation

Education

M.Sc. Mechatronics, Hamburg University of Technology (TUHH), Germany Since Oct 2022

- Specialization: Robotics and Intelligent Systems
- **Courses:** Control Systems, Nonlinear Dynamics, Control Lab, Optimal and Robust Control, Applied Humanoid Robotics, Embedded Systems, Robotics & Navigation in Medicine(Worked with Medical robot- Robotic Needle Placement).

Master Thesis, Hochschule für Angewandte Wissenschaften Hamburg, Germany Sep 2025 - Feb 2025

- Topic: Real-time predictive control for mechatronic systems based on implicit multilinear models.
- Designed and implemented advanced MPC algorithms for implicit multilinear models to test on real-time control.
- Implemented Simulink Real-Time control on inverted pendulum system (Qube-Servo 3)[*Results*] [Manuscript under review at IFAC World Congress 2026] and two-wheeled self-balancing robot (MiniSegway).
- Designed and tuned a PID controller as a baseline and systematically evaluated its performance against MPC.

B.E. Mechanical, R.V. College of Engineering (RVCE), India Aug 2016 - Aug 2020

- Focus Control Systems, Thermal engineering, Aerospace engineering, Machine design, Kinematics of machines, Finite element methods, Mechanical vibrations, 3D Printing, Control engineering, Cutting tool design.

Experience

Research Assistant, Hochschule für Angewandte Wissenschaften Hamburg, Germany Since Feb 2025

- Programming of sensor data acquisition in embedded systems using Python for real-time control applications.
- Applied tensor-train decomposition for efficient identification of high-dimensional dynamical systems, enabling efficient system identification and parameter recovery.
- Developing hierarchical and centralized MPC strategies for building energy management systems.

Research Assistant, Institut für Laser- und Anlagensystemtechnik,TUHH, Germany Mar 2024 – Jan 2025

- Worked on *DigitAI* project, focusing on additive manufacturing of Al alloys. Data analysis and applied ML models (Regression, MLP, CNN, LSTM) for predictive analysis of temperature and melt pool volume.
- Worked on *ILSC-APP* project, focusing on lightweight CAD design. Automated STEP file extraction, performed statistical data analysis, and developed ML-based complexity classification *Git Repo*.
- Explored Graph Neural Networks (GNNs) for CAD classification, improving design optimization workflows *Git Repo*.

Student Tutor, Institute of Control Systems, TUHH - Hamburg, Germany Oct 2023 – Feb 2024

- Tutor for the course Control Systems Theory and Design, Winter Semester 2023/24.

Assistant Mechanical Engineer, S.L.N ENGINEERING WORKS - Bengaluru, India Sep 2021 – Aug 2022

- Worked in Research & Development Department (Automation of Production line).
- Designed and developed over 10 custom work-holding and cutting tools using Siemens NX, reducing machining time by 15% and improving tool lifespan by 5%.
- Managed production line planning and observation, identifying bottlenecks and streamlining operations to reduce downtime.

Publications

- **"Real-time Predictive Control for Mechatronic Systems Based on Implicit Multilinear Models"**, Submitted to *IFAC World Congress 2026* (Under Review).
- **"Tensor Train Based Explicit Multilinear Modeling and Control of Heating Systems"**, Submitted to *IFAC World Congress 2026* (Under Review).
- **"Effect of Laser Parameters on Vibrational Characteristics of AlSi10Mg Parts Produced Using Selective Laser Melting (SLM)"**, *Lasers in Engineering (Old City Publishing)*, Vol. 51, Issue 1–5, pp. 187–203, 2021.
- Presented paper titled **"Enhancement of Heat Transfer Rate of Fin by Varying Fin Parameters"** at the *2nd International Journal of Recent Advances in Manufacturing*.

Projects

Reinforcement Learning based Locomotion and Gait Generation for Petoi Bittle – Institute for Digital and Autonomous Construction, TUHH [Oct 2024 – Feb 2025] *Git Repo* | *Project Report*

- Applied RL techniques (PPO, DDPG, TD3) to train Petoi Bittle for gait generation and step climbing.
- Leveraged simulation environments such as OpenAI Gym and PyBullet.
- Addressed sim-to-real transfer using target joint positions and Petoi Skill Composer for hardware deployment.

Data-Driven Model Predictive Control using Gaussian Processes – Research Project [Nov 2023 – May 2024] *Git Repo* | *Project Report*

- Developed a probabilistic MPC controller using Gaussian process models to address model uncertainty.
- Integrated uncertainty into long-term planning for robust control in dynamic systems.
- Implemented nonlinear solver in MATLAB for improved control performance across test scenarios.

Design and Fabrication of UR-3 Cobot – Research Project [Aug 2017 – Sep 2019]

- Designed and developed dynamic and kinematic models for a collaborative robot (Cobot).
- Implemented trajectory planning for pick-and-place tasks. Developed a PLC-based automation control system.

Bremen Big Data Challenge, Bremen, Germany

- **Multi-Class Classification:** Predicted emotions and context from smartwatch biosignal data. [2024] *Git Repo*
- **Binary Classification** of fraud detection for anti-money laundering using transactional bank data. [2025] *Git Repo*
- Applied preprocessing and data augmentation techniques.
- Trained and fine-tuned models including Random Forest, LSTM, and CNN using TensorFlow.

Skills

Programming: Python, MATLAB, C/C++, Java

Control & Modeling: MPC, Data-Driven Control, System Identification, Optimization-Based Control

Machine Learning: Reinforcement Learning, Neural Networks (MLP, CNN, LSTM, RNN), Supervised Learning

Experimental & Process Systems: Experimental Design, Data Acquisition, Sensor Integration, Real-Time Systems, Process Data Analysis

Frameworks & Tools: PyTorch, TensorFlow, Keras, Scikit-learn, Simulink, ROS2

Mechanical Design & Simulation: SolidWorks, CATIA V5, Siemens NX, Fusion 360 | ANSYS, Adams

Languages: English (C1), German (B1.2 – improving toward C1), Hindi, Kannada

Interests and Hobbies

Drama: Was a active participant in state-level drama competitions; awarded 2nd prize at RANGATHARANGA 2018.

Student Leadership and Volunteering: Member of Dormitory Council, involved in organizing cultural, sports, and social events. [Since 2024]

Cooking and Travel: Enthusiastic home chef and cultural explorer through cuisine and destinations.

Swimming and Piano: Recreational swimmer and basic-level pianist.